

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination: Quiz - 4  
Duration: 30 minutes

Semester: Summer 2025  
Full Marks: 20

**CSE 340: Computer Architecture**

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. Assume that a memory access takes 20 ns if it results in a cache hit, and 50 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

| Index | Valid | Tag | Data         |
|-------|-------|-----|--------------|
| 000   | 1     | 100 | Data[100000] |
| 001   | 0     |     |              |
| 010   | 0     |     |              |
| 011   | 0     |     |              |
| 100   | 1     | 100 | Data[100100] |
| 101   | 0     |     |              |
| 110   | 1     | 011 | Data[011110] |
| 111   | 1     | 111 | Data[111111] |

- a) The size of a block is **8 bits**. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below. [8]

| Memory Address | Hit/Miss | Access Time |
|----------------|----------|-------------|
| 36             |          |             |
| 56             |          |             |
| 26             |          |             |
| 63             |          |             |
| 30             |          |             |

|    |  |  |
|----|--|--|
| 16 |  |  |
| 29 |  |  |
| 37 |  |  |

- b)** Find the bit size of offset bits for the given addresses. [1]

Answer:

- c)** Complete the cache table simulation in the first table given. [3]

- d)** Find the total access time. [1]

Answer:

- e)** Find the AMAT. [1]

Answer:

- f)** Find the hit ratio. [1]

Answer:

- g)** Find the miss ratio. [1]

Answer:

- h)** What is the size of the cache in bits? [1]

Answer:

- i)** Explain how write through works. Is there any optimization problem that occurs if we use write through? If yes, how can we overcome it? [3]

Answer:

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1. Assume that a memory access takes 10 ns if it results in a cache hit, and 40 ns if it results in a cache miss. The current state of the direct-mapped cache is shown below:

| Index | Valid | Tag | Data          |
|-------|-------|-----|---------------|
| 000   | 1     | 100 | Data[1000000] |
| 001   | 0     |     |               |
| 010   | 0     |     |               |
| 011   | 0     |     |               |
| 100   | 1     | 100 | Data[1001000] |
| 101   | 0     |     |               |
| 110   | 1     | 011 | Data[0111100] |
| 111   | 1     | 111 | Data[1111110] |

- a) The size of a block is **16 bits**. Using the table below, calculate the total access time for the given memory sequence using a direct-mapped cache. Indicate hit/miss and access time for each entry. Write your final answers in the table below. [8]

| Memory Address | Hit/Miss | Access Time |
|----------------|----------|-------------|
| 35             |          |             |
| 126            |          |             |
| 24             |          |             |
| 68             |          |             |
| 37             |          |             |
| 18             |          |             |

|    |  |  |
|----|--|--|
| 23 |  |  |
| 34 |  |  |

- b)** Find the bit size of offset bits for the given addresses. [1]

Answer:

- c)** Complete the cache table simulation in the first table given. [3]

- d)** Find the total access time. [1]

Answer:

- e)** Find the AMAT. [1]

Answer:

- f)** Find the hit ratio. [1]

Answer:

- g)** Find the miss ratio. [1]

Answer:

- h)** What is the size of the cache in bits? [1]

Answer:

- i)** Explain how write allocation works. Is there any optimization problem that occurs if we use write allocation? If yes, is there any better way than write allocation to do the same task? [3]

Answer: